

SIMATS ENGINEERING

CO3 - ASSESSMENT TOOL 1

Case Study

COURSE NAME: DEEP LEARNING

COURSE CODE: MLA0408

COURSE OUTCOME COVERED

CO3: Analyze and apply convolutional neural network architectures, including convolutional layers, filters, parameter sharing, regularization, AlexNet and ResNet, to develop solutions for image classification and real-world computer vision applications. (L4)

CO Weightage: 25%

Assessment Tool Weightage: 25%

QUESTIONS:

Total Marks: 25

Case Study 1: CNN for Automated Medical Image Classification

A healthcare organization wants to develop a deep learning system to automatically classify chest X-ray images into **normal** and **disease-affected** categories. The dataset contains thousands of X-ray images with variations in size, brightness and image quality. A CNN model is proposed because it can automatically learn important visual features such as edges, textures, and abnormal regions without manually designing features.

Question:

1. Design and explain a suitable **CNN architecture** for this medical image classification system.
2. Describe the **motivation for using CNNs** and explain the role of **convolutional layers, filters, pooling layers, and fully connected layers** in the proposed architecture.
3. Explain how **parameter sharing** reduces the number of trainable parameters compared with a fully connected network.
4. Discuss suitable **regularization techniques such as dropout, data augmentation and batch normalization** to reduce overfitting.
5. Finally, explain whether **AlexNet or ResNet** would be more appropriate for this application and justify your choice based on feature learning, network depth, computational requirements and classification performance.

Case Study 2: CNN-Based Autonomous Vehicle Object Recognition

An autonomous vehicle company is developing a vision system that must identify **cars, pedestrians, traffic signs, bicycles, and road obstacles** from images captured by cameras mounted on the vehicle. The system must recognize objects under different lighting conditions, viewing angles, distances and road environments. The development team is considering **AlexNet and ResNet** architectures for building the CNN-based recognition system.

Question:

1. Analyze how a **CNN architecture** can be designed for this autonomous vehicle application.
2. Explain the purpose of the **input, convolution, activation, pooling, and fully connected/output layers** and describe how convolutional **filters progressively learn low-level and high-level features**.
3. Explain the importance of **parameter sharing** in reducing computational complexity and improving translation-related feature detection.
4. Discuss the possible problem of **overfitting** and recommend suitable regularization methods.
5. Compare **AlexNet and ResNet** in terms of architecture, depth, parameter efficiency, training difficulty, vanishing-gradient problems and feature-learning capability.
6. Finally, recommend the most suitable architecture for the autonomous vehicle system and justify your decision based on **accuracy, computational cost and real-time application requirements**.

Code of Conduct:

I Haresh Kumar N L (192425009) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: 

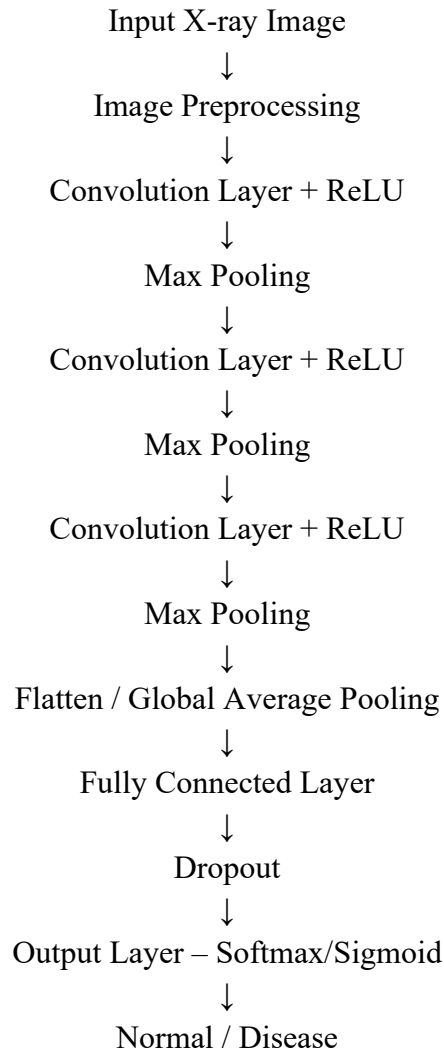
RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding ; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

Answers:

Case Study 1: CNN for Automated Medical Image Classification

1) A suitable CNN architecture can be designed as follows:



Layer	Function
Input	$224 \times 224 \times 1$ grayscale X-ray
Conv2D	32 filters, 3×3 , ReLU
Max Pooling	2×2
Conv2D	64 filters, 3×3 , ReLU
Max Pooling	2×2
Conv2D	128 filters, 3×3 , ReLU
Max Pooling	2×2
Global Average Pooling	Reduces spatial dimensions

Layer	Function
Dense	128 neurons, ReLU
Dropout	0.5
Output	1 neuron, Sigmoid

2) Convolutional Layer

The convolutional layer applies small filters to the X-ray image.

Early layers can detect:

- Edges
- Lines
- Intensity changes
- Simple shapes

Deeper layers can learn:

- Textures
- Lung structures
- Abnormal patterns
- Disease-related regions

Filters

Filters are small matrices, such as 3×3 , that slide across the image. Each filter learns a particular feature during training.

For example:

Input image → **Filter** → **Feature map**

Different filters learn different patterns.

ReLU Activation

The ReLU function is:

$\text{ReLU}(x) = \max(0, x)$

It introduces non-linearity and helps the network learn complex relationships.

Pooling Layer

A **Max Pooling** layer reduces the spatial dimensions of feature maps.

For example:

$2 \times 2 \rightarrow 1 \times 1$

Pooling:

- Reduces computation
- Reduces the number of parameters
- Retains important features
- Provides some tolerance to small image shifts

Fully Connected Layer

The fully connected layer combines the extracted features and uses them to perform classification.

For this case, the final layer can produce:

$P(\text{Disease})$

If the probability is greater than the selected threshold, the image is classified as disease-affected.

3) Parameter sharing is one of the major advantages of CNNs.

In a fully connected network, every input pixel may have a separate weight connected to every neuron. For a large X-ray image, this produces a very large number of parameters.

In a CNN, the same filter is reused across different locations of the image.

For example, a 3×3 filter contains only:

$$3 \times 3 = 9$$

weights, plus a bias.

The same 9 weights are applied throughout the image.

Advantages

- Significantly reduces trainable parameters.
- Reduces memory requirements.
- Reduces computational complexity.
- Allows the same feature to be detected at different locations.
- Helps the network detect translation-related features.

Therefore, CNNs are much more efficient than fully connected networks for image data.

4) Medical image datasets can cause **overfitting**, especially when the model learns the training images too specifically.

The following techniques can reduce overfitting.

a) Dropout

Dropout randomly disables some neurons during training.

For example:

Dropout = 0.5

means approximately 50% of selected neurons are randomly ignored during each training iteration.

This prevents excessive dependence on particular neurons.

b) Data Augmentation

New training examples can be created by modifying existing X-rays.

Examples include:

- Small rotations
- Horizontal/vertical shifts where medically appropriate
- Zooming
- Cropping
- Brightness adjustments

This helps the model become more robust to variations in image quality.

c) Batch Normalization

Batch normalization normalizes intermediate activations during training.

Benefits include:

- More stable training
- Faster convergence
- Improved optimization

- Some regularization effect

d) Early Stopping

Training can be stopped when validation performance stops improving. This prevents the model from continuing to memorize the training dataset.

5)

Feature	AlexNet	ResNet
Architecture	Traditional CNN	Residual CNN
Depth	Relatively shallow	Can be very deep
Feature learning	Good	Excellent
Vanishing gradient	More susceptible	Reduced using skip connections
Training	Easier for smaller models	More complex but effective
Parameter efficiency	Less efficient	Generally better for deeper networks
Computational requirement	Lower	Higher depending on version
Classification performance	Good	Usually stronger for complex images

Recommended: ResNet

For the chest X-ray application, **ResNet** would generally be more appropriate.

The major advantage is the use of **residual/skip connections**:

$$\text{Output} = F(x) + x$$

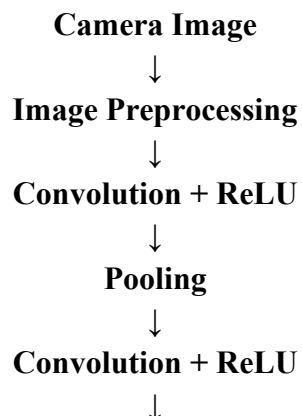
Instead of forcing a layer to learn the complete mapping, the network learns a residual function. This helps information and gradients flow through very deep networks and reduces the vanishing-gradient problem.

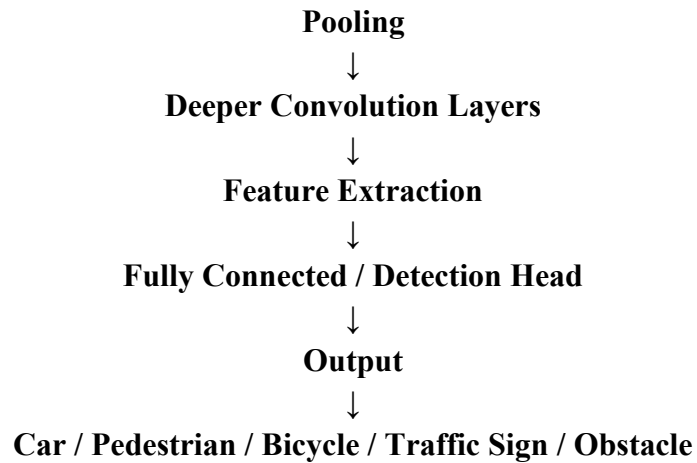
ResNet can therefore learn complex hierarchical features that may be useful for identifying subtle abnormalities in chest X-rays.

However, computational requirements should be considered. A smaller model such as **ResNet-18 or ResNet-34** can be selected when computational resources are limited.

Case Study 2: CNN-Based Autonomous Vehicle Object Recognition

1) A CNN-based object recognition system can be designed as:





For a real autonomous vehicle, object **detection** is more appropriate than simple image classification because the system needs to identify objects and their locations in the scene. A modern implementation could use a ResNet-based backbone with a suitable detection head.

Example architecture:

Stage	Purpose
Input	Camera image
Conv Layer	Detect basic visual patterns
ReLU	Introduce non-linearity
Pooling	Reduce spatial size
Deep Conv Layers	Learn complex object features
Feature Pyramid/Detection Head	Handle different object sizes
Output	Object class + location

2) Input Layer

The input layer receives images from vehicle-mounted cameras. The image is resized and normalized before being given to the CNN.

Convolution Layer

Convolution filters scan the image and create feature maps.

The first layers learn simple features such as:

- Edges
- Corners
- Lines
- Color/intensity patterns

Activation Layer

ReLU introduces non-linearity:

This enables the network to learn complex patterns.

Pooling Layer

Pooling reduces the spatial dimensions of feature maps.

It:

- Reduces computation
- Reduces memory requirements
- Preserves important information
- Makes features somewhat robust to small translations

Fully Connected/Output Layer

The final layers use the learned features to determine the object category.

Possible outputs include:

- Car
- Pedestrian
- Bicycle
- Traffic sign
- Road obstacle

For object detection, the output additionally includes the object's **location/bounding box**.

3) CNN filters use the same parameters at different locations in an image.

For example, a filter that learns the edge of a car can detect that type of edge whether it appears on the left, center or right side of the image.

Benefits

1. Fewer parameters

The same filter weights are reused.

2. Lower computational complexity

There are fewer parameters to store and optimize.

3. Translation-related feature detection

A feature can be detected regardless of its exact position.

4. Better generalization

The network learns general visual patterns rather than memorizing individual pixel positions.

This is particularly important in autonomous vehicles because objects can occur anywhere in the camera frame.

4) The autonomous vehicle dataset can contain thousands of images, but the model may still overfit to specific environments.

For example, it might learn to recognize a particular road background rather than the actual object.

Recommended methods

Data Augmentation

Apply:

- Rotation
- Scaling
- Cropping
- Translation

- Brightness variation
- Contrast variation
- Weather-related transformations where appropriate

This helps the model handle different environmental conditions.

Dropout

Randomly removes neurons during training and reduces dependence on specific neurons.

Batch Normalization

Normalizes activations and helps make training more stable.

Weight Regularization

L1/L2 regularization can discourage excessively large weights.

Early Stopping

Training is stopped when validation performance stops improving.

These methods help the model generalize to unseen road environments.

5)

Factor	AlexNet	ResNet
Architecture	Conventional CNN	Residual CNN
Depth	8 major learned layers	Can have 18, 34, 50, 101+ layers
Skip connections	No	Yes
Vanishing gradients	More susceptible	Significantly reduced
Feature learning	Good	Excellent
Training deep networks	More difficult	Easier
Parameter efficiency	Less efficient for comparable deep models	Better design for deep networks
Computational cost	Lower for original AlexNet	Depends on ResNet version
Complex feature extraction	Good	Very strong
Modern applications	Less commonly preferred	Widely suitable

Vanishing Gradient Problem

In very deep networks, gradients can become extremely small while being propagated backward.

This makes earlier layers difficult to train.

ResNet addresses this using **skip connections**:

$$y = F(x) + x$$

The shortcut allows information and gradients to flow more effectively through the network.

6) Recommended: ResNet

ResNet is the better choice for the autonomous vehicle application.

Reasons

1. Higher accuracy

Autonomous vehicles need to recognize objects reliably. ResNet's deeper architecture can learn more complex visual features.

2. Better feature learning

The network can learn hierarchical representations:

Edges → Shapes → Object parts → Complete objects

For example:

Edges → wheels/windows → vehicle → car

3. Reduced vanishing-gradient problem

Skip connections allow deeper networks to be trained effectively.

4. Robustness

The vehicle must operate under:

- Different lighting
- Different viewing angles
- Different distances
- Different road environments

A deeper CNN can learn more robust features.

5. Real-time consideration

Although ResNet can require more computation than AlexNet, a lightweight variant such as

ResNet-18 can provide a good balance between accuracy and computational cost.

For a production autonomous vehicle, the model can also be optimized using techniques such as:

- Model pruning
- Quantization
- Hardware acceleration
- Smaller ResNet variants